

Lipids

Metabolism

1- Net generation of energy on complete oxidation of a 17-carbon fatty acid is?

Equal to the energy generation from a 16-carbon fatty acid

Equal to the energy generation from an 18-carbon fatty acid

✓ Less than the energy generation from a 16-carbon fatty acid

In between the energy generation from a 16-carbon fatty acid and an 18-carbon fatty acid

Lipids

2- A 45-year-old man has a mild heart attack and is placed on diet and mevastatin therapy. Which of the following will be a result of this therapy?

Low oxidation of fatty acids

Ketosis

✓ Low blood LDLs

High blood cholesterol

3- Diets that have a high polyunsaturated to saturated fatty acids ratio can cause?

Skin lesions

✓ Decrease in serum cholesterol

Increase in serum triglycerides

Decrease in serum HDL

↑ Polyunsat ~ Sat FA ⇒ ↓ Serum cholesterol (Good)
 (causes)

No?, WTF ←

Not starving ←

A course cure
 the stop HMG-CoA
 Really ?? ←

4- Glycerol released from adipose tissue by hydrolysis of triglycerides is mainly?

- Excreted from the body
- Reutilized in adipose tissue
- Taken up by extrahepatic tissues
- ✓ Taken up by liver

lipids

5- What enzyme often malfunctions in diseases associated with the symptoms of steatorrhea (lipids in feces)?

- ✓ Pancreatic lipase
- Lipoprotein lipase
- Phospholipase D ✗
- Acetyl Co-A carboxylase ✗

AA

6- An amino acid that cannot be synthesized from glucose is?

- Serine
- Glycine
- Glutamate
- ✓ Histidine

7- Which one of the following tissues can metabolize glucose, fatty acids, and ketone bodies for ATP production?

- Brain ✓ ✗ ✓
- Muscle ✓ ✗ ✓
- Hepatocytes
- ✓ Muscle
- Liver

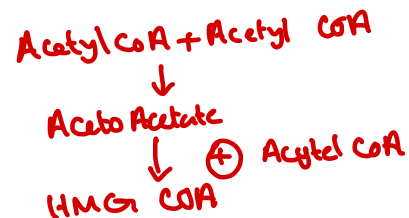
lipids

8- When excess acetyl-CoA, produced by the liver, cannot be used by the TCA cycle, it accumulates in the body as

Acetyl-CoA

Acetoacetate ✓

Glucose

✓ Triglycerides (*fat*)*lipids*

9- Degradation of proteins is facilitated by a process known as

Calcification ✗

Polymerization ✗

✓ Ubiquitination ✓

Esterification

*For Calcium**for F.A. shunt
↓
acid
↓
ions**For FAT, cholesterol
proteins*

THF
Glycine
Glutamate } Participate in
De Novo syn of
Purine ring
except
Ammonia

10- The origins of the nitrogen atoms for *de novo* synthesis of the purine ring are

✓ glutamine, glycine, aspartate

glutamine, glycine

glutamine, glycine, aspartate, ammonia

glutamine, glycine, aspartate, carbamoyl phosphate

11- A fatty acid with 18 carbon atoms will undergo how many cycles of beta oxidation?

✓ 8

6

9

7

$$18 - 1 = 17 \Rightarrow \frac{17}{2} = 8.5$$

(Phenylalanine $\rightarrow$ Tyrosine)

12- Phenylketonuria usually results from a deficiency of phenylalanine hydroxylase or a defect in the regeneration of?

Tetrahydrofolate THF ?

Pyridoxal phosphate

✓ Tetrahydrobiopterin ✓ THb
biopterine

Thiamine pyrophosphate

13- Tyrosine is a non-essential amino acid because we can synthesize it from?

Phenol

Arginine

✓ Phenylalanine

Tryptophan

14- The using of which of the following enzymes will destroy fibrin clots?

✓ Streptokinase

Transaminase

Entrokinase

Asparaginase

15- Inhibition of the synthesis of ATP during oxidative phosphorylation by oligomycin is thought to be due to

- Uncoupling of electron transfer between NADH and flavoprotein
- Blocking of the proton gradient between NADH-Q reductase and QH₂
- Blocking of the proton gradient between cytochrome c₁ and cytochrome c
- ✓ Inhibiting of mitochondrial ATPase (ATP synthase)

16- Which of the following is used by rapidly dividing cells for DNA synthesis

- Arginine
- Histidine
- ✓ Glutamine
- Tyrosine

↓
Glutamine

17- Which statement out of the following is incorrect about Zellweger syndrome?

- Characterized by hypoglycemia and ketosis
- ✓ Caused by a defect in the short chain fatty acid β -oxidation
- Death occurs within 1 years of life
- Results from the absence of functional peroxisomes ✓

←

AA

18- Transamination of amino acids results in products that are called

- Branched chain amino acids ✗
- ✓ α -keto acids ✓
- urea ✗
- β -ketothiolates ✗

Peroxisomes
(doesn't break up
long chain F.A)

Lipids

19- The hyperuricemia associated with disorders such as Lesch-Nyhan syndrome and gout may result from a deficiency of the enzyme?

glutamine:PRPP amidotransferase

glutamine synthetase

✓ ^H ^A ^P ^A ^T
hypoxanthine-guanine phosphoribosyl transferase

PRPP synthetase

HGPRT

HGPRT ✓

Hypoxanthine - Guanine

Lipids

20- Plasma becomes milky?.

✓

After a meal ✓

Due to high level of HDL

During to fasting

Due to high level of LDL

21- The ^{محرر}key regulatory enzyme of cholesterol synthesis is?

(Inhibited by statins)

HMG Co A lyase

HMG- Co A synthase

Mevalonate kinase

✓

HMG Co A reductase

22- Which of the following metabolite integrates glucose and fatty acid metabolism?

Pyruvate

Lactate

✓

Citrate

Acetyl CoA

23- What types of molecules serve as the starting materials for synthesis of uric acid?

Complex lipids such as lecithin and cholesterol

Amino acids

✓ Purine bases

Carbohydrates

24- The using of which of the following enzymes will help DISTINGUISH liver disease and bile duct obstructions from bone disease?

ALP

LD

✓ GGT

ACP

25- OMP decarboxylase is important in?

✓ De novo synthesis of pyrimidines

De novo synthesis of purines

Salvage of purine ..

Salvage of pyrimidine

Alkaline phosphatase
ALP ⇒ liver disease
bone disease

Liver Disease → GGT
+
Bile Ducts Clogage

De Novo syn
of Pyrim

lipids ①

26- Pancreatic juice does not contain which of the following?

Lipase and co lipase ✓

Lipoprotein lipase ✓

Phospholipase A2 ✓

✓ Bile salts → liver juice

27- Conjugation of bilirubin in the liver is required to?

form a complex with serum albumin ?

✓ increase the water solubility of bilirubin ✓

convert bilirubin to a form that no longer absorbs light

convert bilirubin to biliverdin

Carbs

28- Glucagon enhances the release of glucose from liver. Activation of which of the following enzymes facilitates this release?

Glucokinase → (like hexokinase but for liver)

Amylase ✗

Phosphorylase C G

← Glycogen Synthase

29- An uncoupler of oxidative phosphorylation would most likely lead to?

A dependence on ADP for the electron transport chain to operate No

Inability to oxidise NADH by the electron transport chain ??

✓ Operation of the electron transport chain without ATP production

Inability to produce heat by the operation of the electron transport chain ??

Pancreatic

Bile salts are from liver
stored in gallbladder.

Hexokinase for muscle

Amylase for carbs

for glycogen

ADP!! Hell No

Fuck No (we need
to remove H⁺ from NADH)

Handwritten notes:
 (complete)
 → HGPRT
 (hyper uricemia)

30- Sedentary persons who consume rich diets sometimes suffer gout, a condition which may cause toes to swell and joints to become painful due to an accumulation of?

Acetylcholine

Glucose

✓ Uric acid ✓

Glycogen

Gout → uric Acid

بیوشیمی

31- Gluconeogenesis is the production of glucose from non-carbohydrate molecules. Which of the following is not a substrate for gluconeogenesis?

✓ Keton bodies

Lactate

Glycerol from fats

Aspartic acid (AA from Protein)

Carbs

32- All are gluconeogenic key reactions, EXCEPT.....?

Dephosphorylation of glucose 6-phosphate by G6 phosphatase

Conversion of PEP to pyruvate by pyruvate kinase

Decarboxylation of cytosolic oxaloacetate by PEPCK

✓ Carboxylation of pyruvate by PC

33- Pentose phosphate pathway is responsible to make all EXCEPT

- Energy
- NADPH
- Pentose
- Fructose

✓

Carb

34- What is the importance of HMP SHUNT?

✓

- Synthesis of ribose+ NADPH
- Synthesis of fructose+glucose
- Synthesis of glycose+ FADH
- Synthesis of glucose+NADH

Carb

35- Which of the following statements is true about Cori Cycle?

✓

- It is important in the gluconeogenesis pathway
- It involves three tissues: muscle, liver and brain
- It involves transport of lactate from liver to skeletal muscle
- It is active during feeding state

Carb

36- All are allosteric (key enzyme) enzymes in glycolysis pathway, EXCEPT

- Hexokinase ✓
- Pyruvate kinase ✓
- Phosphofructokinase -1 (PFK-1) ✓
- Creatine kinase (CPK)

PFK

Lactate → Pyruvate

✓
Ribose 5 Phosphate + NADPH
[X] → Fructose / glucose
glycose / FADH

from muscle to liver
where it turns lactate
into pyruvate

37- Some patients with G6PD deficiency develop hemolytic anemia if they.....?

Produce low amounts of NADH $\times \rightarrow \uparrow$ NADPH

Ingest fava beans

Are treated with oxidant drug

Contract a severe infection

38- Which of the following enzymes is the key enzyme in pentose phosphate pathway?

Glucose 6 phosphate dehydrogenase

Pentose phosphate epimerase

Aldolase

Pentose phosphate isomerase

39- Which of the following reactions in TCA produces high energy electrons?

citratecis-aconitate

fumaratemalate

cis- aconitateisocitrate

isocitrate α -ketoglutarate

40- In the pathway of Glycolysis, the synthesis of ATP will be catalyzed by.....

Phosphohexose isomerase

Pyruvate kinase

Glucokinase

Aldolase

ADP PEP PK
[ATP] Pyruvate

produce
 $\text{malate} \Rightarrow \uparrow \text{NADHP}$

Glycolysis	HMP shunt
Glucose ↓ G6P	G6P ↓ Glucose

Carb

Pentose ph

Carb

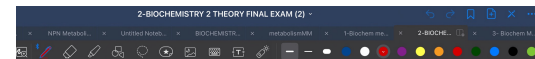
NIAD

← راجع تبدیل فقط، می‌بایستی ATP

يعطي ATP من KDP

← راجِ يَخذ من ATP

$\times \rightarrow \leftarrow$



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آزمون: Biochem

Carb

37- Some patients with G6PD deficiency develop hemolytic anemia if they.....?

Produce low amounts of NADH $\times \rightarrow \uparrow$ NADHP

Ingest fava beans

Are treated with oxidant drug

Contract a severe infection

41- Glucose is transported from blood to the muscle cells by which type of Glucose Transporter?

GLUT-5



GLUT-4

GLUT-2

SGLT-1

42- Which is not a fate for pyruvate? Being converted into.....



Urea

Acetyl Co A

Alanine

Lactate

Carb

Carb

in TCA
in AA making
in lactate
in fermentation

@vip3y